Nicolò Carpentieri

SYSTEMS & COMPUTER ARCHITECTURE

PhD Student, Systems Research Group
Technical University of Munich

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Research Interests

My interests lie in **computer architecture and systems**, particularly **memory disaggregation** at rack scale and the scale-up interconnects, like CXL and related fabrics, that make it possible. Within that space, I focus on **cache coherence protocols** for heterogeneous systems.

Education

→ **Ph.D. in Computer Science** — Oct 2024 -- exp. Oct 2028
Technical University of Munich
Advisor: Prof. Dr. Pramod Bhatotia

→ **M.Sc. in Computer Engineering** — Oct 2021 -- Mar 2024
Politecnico di Torino
Thesis: Designing and Evaluating Mapping of CNN Layers on an Edge-CGRA

→ **B.Sc. in Electronic Engineering** — Oct 2018 -- Sep 2021
Politecnico di Torino

Experience

RESEARCH

→ **Ph.D. Student, Technical University of Munich** — Oct 2024 -- present
Group: Chair of Computer Systems · *Host*: Prof. Dr. Pramod Bhatotia

- **Cache coherence protocols**: automated synthesis of controllers for heterogeneous systems.
- **CXL bridges**: integrating CXL 3.0 cache coherence into arbitrary host architectures.
- **CXL GPUs**: GPUs with CXL 3.0 integration for symmetric cache coherence with host CPUs.

UNDERGRADUATE RESEARCH PROJECT

→ **Visiting Student, Embedded Systems Laboratory (ESL), EPFL** — 2023 -- 2024
Mapping of convolutional neural network algorithms onto CGRA architectures, in collaboration with DAUIN, Politecnico di Torino.

Publications

CONFERENCES

[C1] Synapse: Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers
Nicolò Carpentieri, Anatole Lefort, David Schall, Julian Pritzi, Pramod Bhatotia
IEEE/ACM MICRO 2026 ★ **Distinguished Paper Award Candidate**

[C2] vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures
Anatole Lefort[*], Julian Pritzi[*], **Nicolò Carpentieri**, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia ([*]equal contribution)
ACM ASPLOS 2026 ★ **Best Paper Award**

[C3] C3: CXL Coherence Controllers for Heterogeneous Architectures
Anatole Lefort, David Schall, **Nicolò Carpentieri**, Julian Pritzi, Nicolai Oswald, Pramod Bhatotia
IEEE HPCA 2026

[C4] Performance Evaluation of Acceleration of Convolutional Layers on OpenEdgeCGRA
Nicolò Carpentieri, Juan Sapriza, Davide Schiavone, Daniele Jahier Pagliari, David Atienza, Maurizio Martina, Alessio Burrello
ACM CF 2024

JOURNALS

[J1] Invited extended journal version of *vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures*
Anatole Lefort[*], Julian Pritzi[*], **Nicolò Carpentieri**, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia ([*]equal contribution)
ACM Transactions on Computer Systems (TOCS) forthcoming, 2026

UNDER SUBMISSION

[S1] CXL-GPU: Cache-Coherent Memory Expansion for GPUs over CXL
Under submission 2026

Projects

→ **Synapse** — Generic synthesis of verified, gem5-ready coherence controllers by decoupling protocol, topology, and policy specifications
synapse-coherence.github.io · github.com/NicCarp11/Synapse

Talks

→ **Synapse: Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers** 2026
IEEE/ACM MICRO 2026, Athens, Greece

→ **Automated Synthesis of Verified CXL Bridges for Heterogeneous Architectures** 2026
DFG SPP 2377 "Disruptive Memory Technologies" Annual Meeting, Dresden, Germany

→ **ProtoSLICC: Automated Synthesis of Gem5-based Cache Coherence Controllers** 2025
gem5 Workshop @ ISCA 2025, Tokyo, Japan

Academic Service

→ **Artifact Evaluation Committee** 2026
ACM ASPLOS 2026 · IEEE/ACM MICRO 2026

Teaching Experience

→ Introduction to Software Engineering (EIST) — *Exam Team* 2024--2026
→ Systems Programming — *Course Organizer* 2026--2027

Student Supervision

→ **Disaggregated, Vendor-Generic APUs with CXL** SS 2025
Victor Fritz Trost — B.Sc. Thesis

Honors and Awards

→ **Distinguished Paper Award Candidate** 2026
IEEE/ACM MICRO 2026
Synapse: Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers

→ **Student Travel Grant** 2026
IEEE/ACM MICRO 2026

→ **Best Paper Award** 2026
ACM ASPLOS 2026
vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures

→ **Induction in IEEE-HKN (Mu Nu Chapter)** 2022
Politecnico di Torino

Skills and Interests

→ *Programming languages* C/C++ · Verilog HDL · VHDL · Python · Shell script · LaTeX · Matlab
→ *Development* Unix/Linux · gem5 simulator · FPGA · CUDA · RDMA
→ *Other interests* Skiing · Tennis · Football · Basketball · Padel · Volleyball · Piano · Flute